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ASSOCIATED DETAIL SPECIFICATION

MANUAL, TECHNICAL REPAIR STANDARDS (TRS)
REQUIREMENTS FOR HM&E EQUIPMENT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sets forth the content requirements and instructions for the development of technical repair standards (TRSs) for use in performing class B overhauls to hull, mechanical, and electrical (HM&E), equipment and components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N6946 AREA TMSS
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

SPECIFICATIONS

MILITARY

MIL-M-24784 - Manual, Technical: General Acquisition and Development Requirements.
MIL-M-24784/1 - Manual, Technical Change Package Requirements.
MIL-M-24784/2 - Manual, Technical Revision Requirements.

HANDBOOK

MILITARY

MIL-HDBK-267 - Guide for Selection of Lubricants and Hydraulic Fluids for Use in Shipboard Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

DEPARTMENT OF DEFENSE

DFAR Sup 27.475-1 - Defense Federal Acquisition Regulation: Technical Data, Other Data, Computer Software, and Copyrights.
DoD 5010.12-L - Acquisition Management Systems and Data Requirements Control List (AMSDL).
DoD 5200.1-R - Information Security Program Regulation.
DoD 5220.22-M - Industrial Security Manual for Safeguarding Classified Information.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications), the text of this specification shall take precedence. When a generic requirement is in conflict with a unique requirement from the governing associated detail specification, the unique requirement shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Security classifications. The overall classification assigned to a technical manual shall agree with the highest classification assigned to any portion within and shall be placed conspicuously at the top and bottom of the Cover and Title page or abbreviated title. Security classification markings shall be in accordance with DoD 5220.22-M and 5200.1-R and shall be placed in accordance with DoD 5220.22-M and 5200.1-R. The security markings shall be the same as for other title pages.

3.2 Deliverable products and data items. The deliverable products and data items of the HM&E TRS shall be specified in the Contract Data Requirements List (CDRL) and shall conform to the requirements specified in MIL-M-24784 and herein. Data items shall include the following, as applicable:

(a) Outline or book plan.
(b) Review draft copy (RDC).
(c) Preliminary technical manual (PTM).
(d) Final reproducible copy (FRC).
(e) Supplements.
(f) Changes (see 3.2.1).
(g) Revisions (see 3.2.2).
(h) Management data (see 3.2.3).
(i) Engineering judgement record (see MIL-M-24784).
(j) Cost and pricing data (see MIL-M-24784).

3.2.1 Change packages. When specified (see 6.2), change packages shall be in accordance with MIL-M-24784/1.

3.2.2 Revisions. When specified (see 6.2), revisions shall be in accordance with MIL-M-24784/2.

3.2.3 Management data. Unless otherwise specified (see 6.2), the following quality assurance data items shall be provided in accordance with MIL-M-24784:

(a) Technical Manual Quality Assurance Program Plan.
(b) Validation Plan.
(c) Validation Certification.
(d) Technical Manual Evaluation Records.
(e) Verification Plan.
(f) Technical Manual Verification Incorporation Certification.

3.3 Arrangement. Unless otherwise specified, the manual shall be arranged in a standardized format (that is, front matter, technical content, appendices, glossaries, indexes and back matter) and appropriately divided by volume, part, chapter and section in accordance with the requirements of MIL-M-24784, Appendix C and the following.

3.3.1 TRS subject matter. The TRS shall provide detailed instructions for performing a class B overhaul. The TRS shall discuss critical procedures (see 6.5.3), testing and maintenance examinations, supporting illustrations, lists of items to be replaced and any special tools needed for the overhaul. The TRS shall

also describe critical operations and checks that must be done during item reassembly, minimum quality control requirements, and specific post-overhaul inspections. The technical content for systems and equipment shall be formatted to meet HM&E (see 3.5) requirements. The TRS sequence and breakdown of divisions shall be as follows:

(a) The technical content shall follow the front matter and shall be arranged in chapters. Each chapter shall consist of one or more sections that are divided into main paragraphs, and then subdivided into subparagraphs. Each section and major (numbered) paragraph shall have a brief title describing the content or action required. The level of detail shall be geared to the journeyman-level mechanic.

(b) Technical content for a compound (composite) item, with major subassemblies (see 6.5.7), shall be divided so that each major item will be addressed in a separate chapter to promote the work-unit, work-package concept (see 6.5.8).

(c) Minor assemblies shall be divided into separate sections and included with their major subassemblies, to promote work group breakdown structure of authorized overhaul facilities.

3.3.2 Multipart volumes. The TRS shall be divided into two parts. Part 1 shall contain all overhaul procedures, requirements, and drawings necessary to perform the overhaul. Part 2 shall record all readings specified in part 1.

3.4 Format and development instructions.

3.4.1 Writing style. Unless otherwise specified, the style of writing, level of writing, readability, and referencing shall be in accordance with the requirements of MIL-M-24784, Appendix B.

3.4.2 Safety precautions. Unless otherwise specified, safety and health warnings, cautions and notes shall be in accordance with the requirements of MIL-M-24784, Appendix D.

3.4.3 Tabular material. Unless otherwise specified, the development of tables shall be in accordance with the requirements of MIL-M-24784, Appendix E.

3.4.4 Graphics. Unless otherwise specified, the development of illustrations, drawings, diagrams and sketches shall be in accordance with the requirements of MIL-M-24784, Appendix F.

3.4.5 Numbering. Unless otherwise specified, the numbering of paragraphs, procedures, divisions, issues, changes, illustrations, tables, and publications shall be in accordance with the requirements of MIL-M-24784, Appendix G.

3.5 HM&E TRS technical content. The TRS shall be developed as a two-volume package. HM&E TRS technical data shall be formatted as follows.

(a) Part one of the volume shall contain:

(1) Title page (see MIL-M-24784).
(2) Front matter (see MIL-M-24784).

(3) Chapter 1 - General information (see 3.5.1).
(4) Chapter 2 - Shipboard removal (and replacement) (see 3.5.2).
(5) Chapter 3 - Unit disassembly (see 3.5.3).
(6) Chapters 4 through "n" - Item overhaul procedures for each major component and assembly with a separate chapter for each component (pump, turbine) and major assembly (see 6.5.2) (governor, governor valve and servomotor, attached oil pump, motor driven oil pump) shall be provided. Each chapter shall discuss disassembly, inspection, repair, reassembly, testing, and so forth (see 3.5.4).
(7) Chapter n + 1 - Unit assembly (see 3.5.5).
(8) Chapter n + 2 - Unit shop test (see 3.5.6).
(9) Chapter n + 3 - Shipboard installation, alignment, and testing (see 3.5.7).
(10) Appendix A - Mandatory overhaul replacement parts list (see 3.5.8.1).
(11) Appendix B - Contingency material list (see 3.5.8.2).
(12) Technical manual deficiency evaluation report (TMDER)/User activity technical manual comment sheet (UATMCS) (see MIL-M-24784).

(b) Part two of the volume shall contain:

(1) Title page (see MIL-M-24784).
(2) Front matter that describes content and instructions for collecting and disposing of the data package (see MIL-M-24784).
(3) Material condition/overhaul report (see 3.5.8.3).
(4) Appendix C - examination, test, and repair action record (ETRAR) (see 3.5.8.4).
(5) Additional appendices (D and on) - special requirements (see 3.5.8.5).
(6) TMDERs (see MIL-M-24784).

3.5.1 Chapter 1 - General information. This chapter shall provide a general overview of the data presented in the TRS. The chapter shall include the following information and identify with numbered paragraph headings as applicable.

3.5.1.1 Scope. The scope shall include a brief summary of the purpose, content, and arrangement of the TRS, including appendices. It shall also state that all commercial and Government activities involved in the overhaul process must meet all TRS class B overhaul requirements.

3.5.1.2 Points of contact. This paragraph shall state that the TRS takes precedence over all other technical documents for overhaul of the commodity. It shall further state that the shop supervisor or design division should be contacted regarding problems associated with use of the TRS.

3.5.1.3 Overhaul conformance. This paragraph shall state that compliance with the applicable chapters of the TRS is mandatory and that, upon completion of the overhaul, equipment must meet the stated performance specifications. This paragraph shall also contain the following information:

(a) Specify that configuration changes are not authorized without Government approval.
(b) Address visual inspection requirements of replacement parts and specify that the TRS does not require replacement part testing.
(c) State that parts listed in the mandatory overhaul replacement parts list (Appendix A - see 3.5.8.1) are included because of high past failure rates and/or cost to repair or replace and that these parts shall be replaced regardless of condition. Therefore, this material shall be on hand prior to performing a class B overhaul in accordance with the TRS.
(d) Specify that the contingency material list (Appendix B - see 3.5.8.2) contains a list of those parts that may need to be replaced during the overhaul. The decision to buy these contingency parts shall be made by the overhauling activity.
(e) Direct the TRS user to record specified readings, test results, inspection and examination results, as well as corrective actions; and forward this data to the TRS maintenance activity.
(f) Specify that data readings are recorded at certain inspection and verification (I&V), see 6.5.5, points during the overhaul. An individual shall be chosen to sign off on all mandatory I&V points. The local overhauling activity may designate additional I&V points if needed.

3.5.1.4 Documentation and references. A tabular listing shall be developed to present all documents, publications, product engineering drawings and schematics, specifications and standards, engineering change data, and other reference data used in the development of the TRS and referenced in the text. Applicable documents shall be listed in the order they are referenced in the text.

3.5.1.5 Unit assembly and subassembly. This paragraph shall identify the repairable items (see 6.5.6) covered by the TRS. Items shall be identified by nomenclature, model number, allowance parts list (APL), commercial and Government entity code (CAGEC), and manufacturer's part number. These data shall be listed in tabular form. This paragraph shall also provide a tabular list of the ship or machinery alterations which have been incorporated into the TRS and the hull numbers to which they apply. If no ship or machinery alterations have been incorporated into the TRS, a statement to that effect shall be included.

3.5.1.6 Parts replacement. Subparagraphs in this topic area shall be developed to specify that:

(a) The local manufacture or fabrication of certain parts is strictly forbidden. Because of the critical nature and stringent manufacturing requirements, the Government has restricted the manufacture of these parts to qualified sources. A tabular listing of all such parts shall be provided. The list shall contain the restricted part by name and part number. In addition, these parts shall be highlighted by an appropriate note in Appendices A or B (see 3.5.8.1 and 3.5.8.2), as applicable. A printout of those parts which have been restricted may be obtained from the Government activity providing supply support for the commodity.

(b) Parts examined which do not meet acceptance criteria may be repaired [if not on Appendix A (see 3.5.8.1) or a restricted part] or shall be turned in to the supply system or disposed of using current procedures and documentation. Local manufacture of

delayed; however, these parts shall conform to drawing specifications and source inspections.

(c) Shop store items such as gaskets and packing material, locknuts and lockwashers, cotter pins, keys, and so forth, are not listed in the mandatory overhaul replacement parts list or contingency material list but shall be replaced. This material shall be on hand to accomplish the planned overhaul maintenance.

3.5.1.7 Special facilities, test equipment, skills, and test conditions. This paragraph shall note the special or unusual equipment, facilities, or personnel skills required to accomplish the TRS equipment overhaul. It shall include the following information as required:

(a) The special facilities required to perform the overhaul shall be identified as prescribed and listed in tabular form.

(b) Special test equipment (or acceptable alternative items) and repair aids required (specify quantities) in the examination, repair (including alignment, adjustment, and troubleshooting), and performance or acceptance testing of the items covered by the TRS, shall be identified and described. A tabular listing is preferred.

(c) Skills requiring special training shall be defined. This information shall be presented in a brief narrative form or a tabular listing.

(d) Special work station environmental conditions required in the overhaul process shall be specified; for example: atmospheric pressure, specific temperatures, relative humidity, or clean rooms. This information may be appropriately detailed in table format.

3.5.1.8 User's comments procedure. A brief instruction to the manual user shall specify that comments or recommendations concerning the manual are to be forwarded for consideration (see MIL-M-24784).

3.5.2 Chapter 2 - Shipboard removal (and replacement). This chapter shall include procedural, inspection, repair, data recording, and testing information with supporting appendices, warning and caution statements, and sketches in accordance with applicable TRS requirements:

(a) Unit boundary identification (addressing overhaul considerations for complete or partial unit overhaul/removal).

(b) Draining unit.

(c) Insulation removal (with appropriate asbestos hazard warnings).

(d) Interference, interface, and piping removal (with appropriate asbestos hazard warnings).

(e) Unit installation site (foundation, interface, and so forth).

(1) Preparation.
(2) Cleaning.
(3) Inspection.
(4) Repair.
(5) Blanking-off, tagging, and so forth (including applicable tag out requirements).
(6) Removal (and replacement) routes (where specific routes have been developed, they shall be included or referenced as required).

3.5.3 Chapter 3 - Unit disassembly. This chapter of the TRS shall specify the requirements for disassembly of the major components of the unit to facilitate the work breakdown structure of a compound unit, as broken down into repair sections in Chapters 4 through "n", as required. Procedural and data recording requirements with supporting appendix pages, warning and caution statements, and sketches in accordance with the applicable requirements shall be provided for the following:

(a) Major unit boundary identification.
(b) Coupling separation/interface and piping removal, electrical disconnection.
(c) Major unit removal.

3.5.4 Chapter 4 through "n" (and others as required) - Item overhaul procedures. These chapters shall contain procedures, minimum test and inspection requirements, processes, methods, and data deemed necessary for the satisfactory accomplishment of a class B overhaul. Acceptance or rejection criteria for wear, deterioration, dimensions, operating parameters, alignment, and other standards of acceptance shall be specified, with limits listed in the text and in Appendix C (see 3.5.8.4). Each component or major assembly shall be defined in a separate chapter. The chapters and the sections within each chapter shall be arranged in a top-down logical sequence of disassembly, repair, and assembly. Each chapter shall contain the following sections:

(a) Removal, disassembly, inspection, and repair procedures.
(b) Examination and acceptance criteria, corrective action, mandatory replacement parts, cleaning methods, and data collection.
(c) Reassembly procedures.
(d) Acceptance testing (if applicable). Subsections will usually discuss subassemblies or individual parts (may be more than one related part per subsection).

3.5.4.1 Removal, disassembly, inspection, and repair procedures. Step-by-step procedures for the disassembly, repair, and inspection of the repairable items shall be developed. Detailed instructions shall be limited to areas where the sequence of operations is not obvious or where special techniques are required for access to a part requiring repair. When the sequence of disassembly is obvious, it is sufficient to state "Disassemble by removing parts in accordance with the disassembly sequence numbers." Additional instructions, cautions, and warnings should be specified in sequence with a disassembly instruction as required. Instructions on the tagging, labeling, and match-marking of parts shall

be included in this paragraph. When examinations and inspections must be performed prior to or during disassembly, the TRS shall so specify. Step-by-step procedures shall be provided in any of the following cases (see MIL-M-24784):

(a) When accomplishment of the required action would not be obvious to a journeyman mechanic unfamiliar with the particular equipment covered by the TRS.
(b) When a peculiar or special action is required.
(c) When safety related or precautionary procedures are required to prevent personnel injury or equipment damage.
(d) When the required action involves an area with a known history of failure due to improper procedures.
(e) When because of the complexity of a particular procedure it is necessary to maintain the continuity of the TRS.

3.5.4.2 Mandatory replacement parts. When the requirement exists for the planned mandatory replacement of particular items, the TRS shall reference the mandatory overhaul replacement parts list (see figure 7-1) - see 6.2. The list shall be in tabular format, in Appendix A, and shall identify those parts that are required to be replaced, regardless of their condition. Replacement materials shall be available at the overhaul activity in accordance with the TRS instructions. Examples of mandatory replacement items may be parts such as seals, washers, vacuum tubes, air filters, and motor brushes.

3.5.4.3 Examination criteria. Criteria to determine the suitability of worn or degraded critical parts for continued use shall be developed and shall explicitly state the measurable wear and deterioration limits which, if exceeded, would require corrective action to maintain reliability and specified performance. These data shall appear in the text and be presented, in tabular form, in Appendix C (see 3.5.8.4). Data shall identify the part and measurable limits. Departures from existing limits shall be highlighted with a text footnote. Corrective actions shall also be specified in the text.

3.5.4.4 Special cleaning methods. When applicable to the repairable items covered in the TRS, special processes and methods of cleaning and examination to be performed on each disassembled item shall be described. Special cleaning and inspection instructions shall include:

(a) The pre-cleaning examination requirements and cleaning process to be used for each item requiring cleaning.
(b) Step-by-step procedures for accomplishing the cleaning process.
(c) Specific identification of cleaning materials to be used by their commonly known name and specification number. These data may be provided in tabular format.
(d) Quantitative instructions for the variables associated with the cleaning process; for example: air pressure, moisture content, solvent temperature, soak time, ultrasonic frequency, drying time, and temperature.
(e) Procedures for use of special tools, jigs, or fixtures.
(f) Post-cleaning examination, preservation, and handling instructions.
(g) Cautions or warnings to be observed to protect personnel and equipment.
(h) Other data as may be determined appropriate for the repairable items.

3.5.4.5 Inspection, data collection, and corrective actions. Procedures, special instructions, examination, and technical information (including supplementary illustrations and tabular lists) required to perform effective inspection and repair of the repairable items covered in the TRS shall be identified (see 6.2). The actions to be performed shall ensure that the repairable items meet or exceed the specified performance requirements of the final acceptance test (see 6.5.4) and satisfies all quality assurance inspection requirements.

3.5.4.5.1 Fault identification. Detailed step-by-step procedures for identifying problems shall be developed.

3.5.4.5.2 Corrective action. Detailed step-by-step procedures for corrective actions, including the repair and replacement of defective, damaged, or deteriorated parts, shall be developed.

3.5.4.5.3 Inspection and verification (I&V). I&V points shall be noted in the TRS left-hand margin by "I" or "V", both in the text and Appendix C. I&V points shall be determined by using the best engineering judgement. All data to be collected shall be specified in the text. The requirement to sign all I&V points shall be included in the front matter of parts 1 and 2 of the TRS. Signature blocks shall be provided as appropriate in the examination, test, and repair action record (ETRAR) (see 3.5.8.4).

3.5.4.5.4 Alignment and adjustment procedures. Detailed procedures shall be identified or developed for sophisticated mechanical and electrical alignments and adjustments. Procedures shall describe the methods used and special tools and equipment required to achieve the setting and positioning of adjustments and controls, and mechanical, gearing, and linkage alignments within the repairable item.

3.5.4.6 Reassembly. Illustrations, reassembly instructions, and examination requirements for the reassembly of the repairable items shall be developed as required. The use of any special tools, jigs, fixtures, or test equipment shall be specified. Where sequence of reassembly is obvious, it is sufficient to state "Reassemble by installing parts in accordance with the reassembly sequence numbers". Additional instructions, cautions, and warnings shall be specified in sequence within a reassembly instruction when required. Figures prepared to support disassembly instructions may be repeated; however, sequence numbers shall be changed to reflect proper numerical order for reassembly. The instructions or procedures shall indicate the following:

(a) The use of corrosion-preventive compounds, paints, or other materials and the use of gaskets or sealing compound materials shall be specified by nomenclature, trade name, specification number, and national stock number (NSN). Non-NSN consumables and materials shall be avoided whenever possible.
(b) Identification of all points requiring lubrication, the kind of lubricants required, and the method of application. Lubricants to be used shall be those specified in the standard lubricants lists. A NSN shall be listed if available. Additional information regarding lubrication can be found in MIL-HDBK-267.

(c) Any special instructions for installing fasteners, safety wiring, cotter pins, and other locking devices.
(d) Special instructions for aligning, adjusting, and measuring tolerances, clearances, end play, backlash, and tolerances between gears and linkage alignment data.
(e) Identification of critical examinations and tests during reassembly and, if required, provisions for data collection.
(f) Reference to tables, appendices, or illustrations as applicable.
(g) Other data as may be determined appropriate for the repairable items type.

3.5.4.7 Acceptance test. Acceptance testing shall be specified for each individual component, if needed, and for the system as a whole. The section shall specify that:

(a) After completing all repair or overhaul actions and when the repairable item is considered ready for final examination, the item shall be inspected for proper identification and tested to verify satisfactory performance.
(b) The acceptance test (see 6.5.1) shall be accomplished in accordance with the established performance requirements. Reference shall be made to pertinent data contained elsewhere in the TRS.
(c) Specified data collection requirements for the item shall be included in the TRS ETRAR (see 3.5.8.4).

3.5.5 Chapter n + 1 (or as applicable) - Unit assembly. This chapter shall specify all criteria addressed in Chapter 3 as applicable to unit reassembly. The following items shall be covered:

(a) Major unit installation.
(b) Piping, interface installation and alignment.
(c) Coupling cold alignment.

3.5.6 Chapter n + 2 (or as applicable) - Shop unit test. This chapter shall be developed to facilitate work breakout for a complete unit that requires a unit shop test prior to shipboard installation. An approved outline for the generic unit shall be used in determining the following unit test requirements:

(a) Test stand requirements.
(b) Unit flush.
(c) Unit shop test requirements.

3.5.7 Chapter n + 3 (or as applicable) - Shipboard installation, alignment and testing. This chapter shall specify requirements for:

(a) Unit installation.
(b) Insulation installation.
(c) Cold alignment recheck.
(d) Piping, interface installation and alignment.
(e) System check.
(f) Coupling hot alignment.
(g) Unit break-in.
(h) Final acceptance test.
(i) Interfacing system inspection.

3.5.8 Appendices. The TRS shall provide pertinent supplementary and reference data required in the inspection, parts replacement, testing, and final acceptance of the repairable items covered in the TRS. Each appendix shall contain a statement delineating its purpose and application. Appendices shall be in accordance with MIL-M-24784.

3.5.8.1 Appendix A - Mandatory overhaul replacement parts list. This appendix shall contain a comprehensive list of items which shall be replaced, regardless of observed condition (see 3.5.4.2). There shall be a separate sheet listing the mandatory replacement parts for each unit subassembly or major subassembly component. The parts designated for the mandatory overhaul replacement parts list shall be specified by the Government acquisition activity using the following considerations:

(a) Parts which are usually damaged or destroyed during equipment disassembly.
(b) Parts normally subject to accelerated wear.
(c) Parts known to be more economical to replace than inspect.
(d) Parts which have a history of high usage.
(e) Parts that have a high documented failure rate and are costly to replace between overhauls. No direct or indirect citation of the actual failure rate of any part is allowed. Technical manuals shall not list or cite any figures of merit for the reliability or maintainability of the equipment (or parts) at any level of indenture.

NOTE: All TRS written requirements shall be met before a class B overhaul will be certified as complete. If a material ordering guide (MOG) exists for an item being covered in the TRS, the MOG may be used to develop appendix A.

3.5.8.2 Appendix B - Contingency material list. A contingency material list shall be developed in accordance with MIL-M-24784. It shall be a tabular listing of parts, similar to figure 7-2. When developed, there shall be a separate sheet for each unit subassembly or major subassembly or component. The contingency material list is a guide for material that may be needed when the unit is overhauled. It shall list the parts which may require replacement during the overhaul of the unit. Any part listed in the contingency material list shall be addressed in the appropriate inspection and repair subsection of the TRS. Those parts whose costs approach or exceed the total cost of the unit shall not be listed. Parts which rarely require replacement and are not normally replacement parts for unit overhaul shall not be listed. A MOG, if it exists, may be used to develop Appendix B.

3.5.8.3 Material condition/overhaul report (MC/OR). A certification document shall state that the overhaul has been properly completed and meets all means specified in the acceptance criteria. It will also provide a means to collect additional information from the line mechanic concerning conditions observed prior to, and during, the overhaul. Directions for recording data and distribution of the report shall be specified in the TRS text. A typical MC/OR is shown in figure 7-3. At a minimum, MC/ORs shall conform to the following:

(a) Check-the-box and short fill-in questions shall be used to the greatest degree possible.
(b) A heading identifying the component shall be completed.
(c) Specific questions regarding the "as found" material condition of the item shall be prepared.
(d) Provision shall be made for the mechanic to supply informal comments and short remarks.
(e) Specific questions or statements regarding the "as overhauled" state of the item shall be prepared.
(f) Provision shall be made for the mechanic to record or highlight any approved or unapproved deviations and out-of-spec data.
(g) A separate MC/OR shall be prepared for each subassembly and major subassembly component and shall be included in the appropriate data section in part 2.
(h) MC/ORs shall be identified in the table of contents, list of effective pages, and TRS text.

3.5.8.4 Appendix C (part 2) - ETRAR. Required ETRAR information (see figure 7-4) shall include the system test step number and the desired response and tolerance limits. Other information may vary according to the equipment type or user requirements. Separate summary sheets shall be developed for each subassembly and major subassembly or component as previously specified in the TRS and assigned an appropriate title. Each instruction to record data in the TRS text shall also be entered in the ETRAR. When possible, a line drawing identifying the measurement to be taken shall be provided on the same page. Reading specifications or parameters that depart from original design characteristics shall be highlighted and footnoted with rationale. A space for signatures at specified data entry and I & V points shall also be provided. The last section of Appendix C for each major subassembly component and the last section of the total unit shop test and unit final acceptance test shall specify the minimum criteria for determining whether the class B overhaul requirements are in accordance with Government testing criteria. No space for recording data on these pages shall be provided. However, activities shall be directed to submit all local shipyard test forms with the completed overhaul record (part 2 of the TRS).

3.5.8.5 Additional appendices (D and on). When specified (see 6.2), the text in the body of the TRS shall specify any other special requirements covered, with additional appendices.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality control. Unless otherwise specified (see 6.2), the quality conformance inspection, Government inspection, validation, verification, technical manual quality assurance (TMQA) program reviews, technical manual evaluation records and rejection requirements shall be in accordance with MIL-M-24784.

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified (see 6.2), the packaging and preservation requirements for delivery of book plans, review draft copies, preliminary technical manuals, final reproducible copy, technical manuals, replenishment materials, changes and revisions shall be in accordance with MIL-M-24784.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The TRS supplements the technical maintenance section of the technical manual for the HM&E item and provides all necessary technical information required to rebuild or restore the HM&E item. Normal maintenance and test procedures covered in the technical manual for the item should not be repeated in the TRS except when deemed necessary for clarity or continuity. Instead, reference to the manual will be made.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification (or any TMCR referencing this specification).
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
(c) Applicable publication number, classification, and distribution statement (see 3.1).
(d) An outline or book plan manual must be specified whenever a technical manual is required for a new equipment or system (see 3.2a).
(e) Double column text must normally be specified. When increased effectiveness of presentation results, a single column format is also acceptable.

(f) A review draft copy (RDC) must always be specified for every new technical manual (see 3.2b). An RDC also must be specified for manual change or revision procurements (see 3.2f and 3.2g), unless changes to the technical manual have been pre-approved (TMDER, ECP, and so forth) by the Government.
(g) Specify the number of weeks required for Government review of technical manual deliverables.
(h) Specify a preliminary technical manual when an extended period of use is required prior to Government verification (normally applies to complex equipment and systems only), see 3.2c.
(i) Final reproducible copy (see 3.2d).
(j) Supplements to manuals (see 3.2e) must be specified when:

(1) Classified information can be confined to the supplement manual such that its basic technical manual will be routed as "unclassified".
(2) It augments a technical manual to provide for a different model and can be justified as a cost effective method of promulgation.
(3) It can be justified as a practical method of promulgation.

(k) Changes must be specified only when they pertain to technical manual corrections resulting from a hardware change or modification. They may also be specified if it has been determined that significant omissions of technical data or information has been identified. Do not specify a change to correct minor or editorial corrections. When changes are justified on the above basis, the change package must include the correction of all outstanding temporary or interim changes and may include minor changes (see 3.2f). The change package must be developed in accordance with MIL-M-24784/1.
(l) Whenever the number of change pages is estimated to exceed approximately 25 percent of the total number of pages in the manual, an update revision must be specified (see 3.2g) and developed in accordance with MIL-M-24784/2.
(m) When the total number of pages requiring change exceed approximately 50 percent of the total number of pages in the manual, a complete revision must be specified (see 3.2g) and developed in accordance with MIL-M-24784/2.
(n) Engineering judgement records (see 3.2i).
(o) Cost and pricing data (see 3.2j).
(p) What quality assurance data items are to be delivered (see 3.2.3, 6.4, and MIL-M-24784).

(1) TMQA program plan.
(2) Validation plan.
(3) Validation certification.
(4) Evaluation records.
(5) Verification plan.
(6) Verification incorporation certification.

(q) Requirements for parts lists (see 3.5.4.2).

(r) Identification of inspection and repair of items covered in the TRS (see 3.5.4.5).
(s) When preservation, packing, and marking are other than as specified (see Section 5. PACKAGING).
(t) For classified equipment, attach a DD Form 254 specifying the following:

(1) Security classification.
(2) Downgrading and declassification notification.
(3) Areas requiring security protection.

6.3 Technical manual acquisition. This specification (or a TMCR based on this specification) must be listed on the Contract Data Requirements List (DD Form 1423) in order to acquire the technical manuals described by this specification, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
3.2.3	DI-M-2194	Manual, Technical; Quality Assurance Program Plan
3.2.3	DI-M-2195	Manual, Technical; Validation Plan
3.2.3	DI-M-2196	Manual, Technical; Validation Certification
3.2.3	DI-M-2197	Manual, Technical, Evaluation Record
3.2.3	DI-M-2198	Manual, Technical; Verification Plan
3.2.3	DI-M-2201	Technical Manual Verification Incorporation Certification

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.5. Definitions. The words or phrases used throughout this specification must be as defined in MIL-M-24784 and the following.

6.5.1 Acceptance tests. An examination and test of an item after completion of its overhaul or repair to ensure that the item satisfies minimum specified performance requirements.

6.5.2 Assembly. A number of parts or subassemblies or any combination thereof joined together to perform a specific function and capable of being disassembled.

6.5.3 Critical procedures. Those procedures which:

(a) Are necessary to restore the equipment to performance specifications.
(b) If not accomplished, would prevent the equipment from operating between overhauls.
(c) Contain safety related or precautionary measures designed to minimize personal injury or equipment damage.

6.5.4 Final acceptance tests (FAT). The examination and testing of repairable items after completion of overhaul or repair to ensure that the items satisfy performance requirements.

6.5.5 Inspection and verification (I&V). Inspection requirements are developed for processes that require either specialized training and certification or independent observation. Verification requirements are developed for processes that do not require independent observation. I&V requirements are identified by the shipyard technical authority to ensure that all repair, overhaul, conversion, or refurbishment of Naval ship systems, subsystems, and components fully meet Navy requirements. TRSs also state when I&V functions are to be performed. Those time periods are called I&V designated points.

6.5.6 Repairable item. An item of durable nature which, when unserviceable, normally can be economically restored to a serviceable condition through repair procedures performed by a Government or commercial overhaul facility.

6.5.7 Subassembly. A portion of an assembly or unit which is replaceable as a whole, but also has a part or parts which are individually replaceable. (The distinction between an assembly and a subassembly is determined by the individual application. An assembly in one instance may be a subassembly in another, where it forms a portion of an assembly.)

6.5.8 Work-unit/work-package concept. An individual unit of information containing all data necessary for a technician to perform a specific task with minimal referencing.

6.6 Subject term (key word listing).

Examination test and repair action record.
Material condition/overhaul report.
Material deficiency evaluation report.
Technical manual identification number.

Review activity:
Navy - EC

Preparing activity:
Navy - SH
(Project TMSS-N242-07)

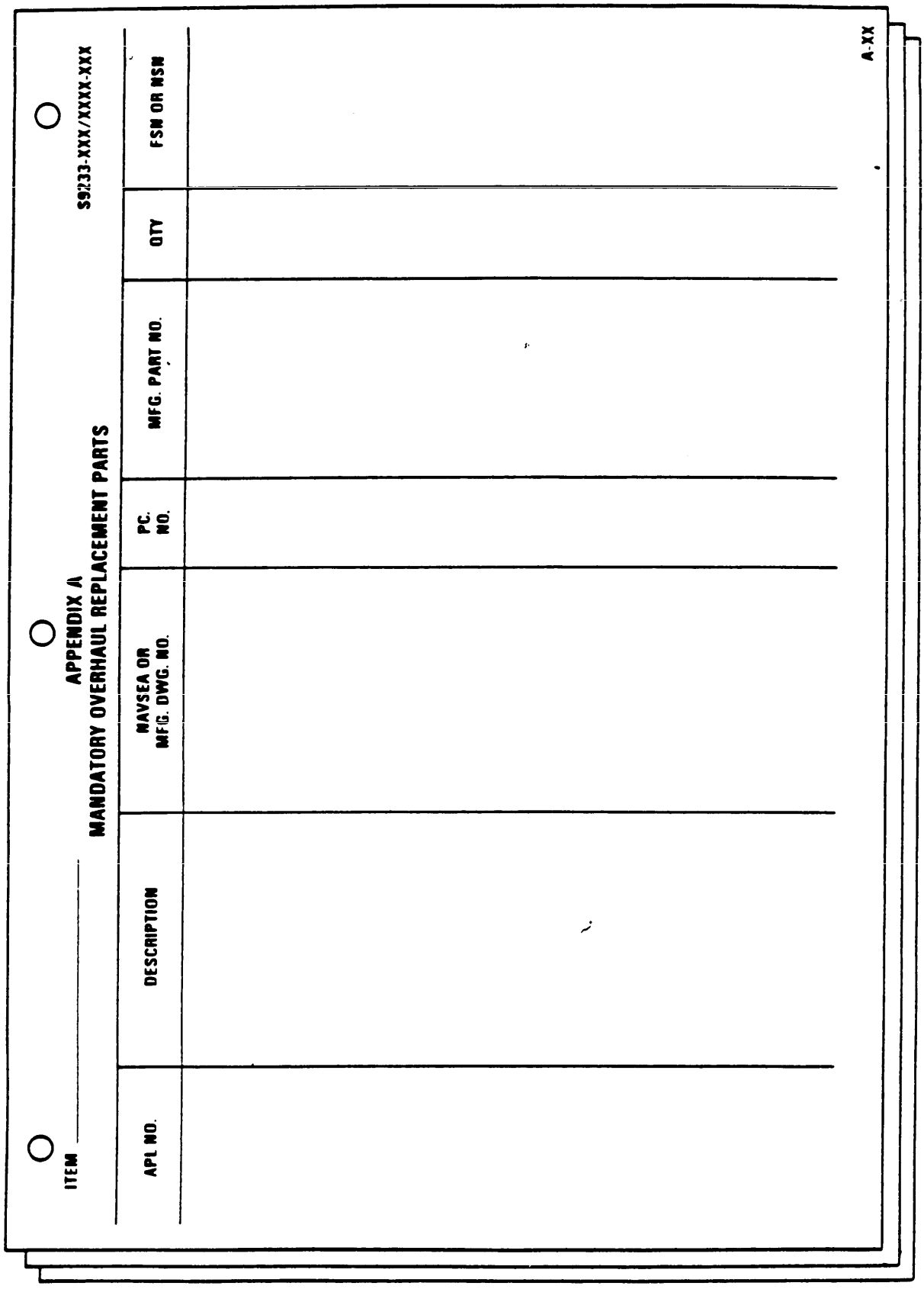


FIGURE 7-1. Overhaul replacement parts list, mandatory. (Sample)

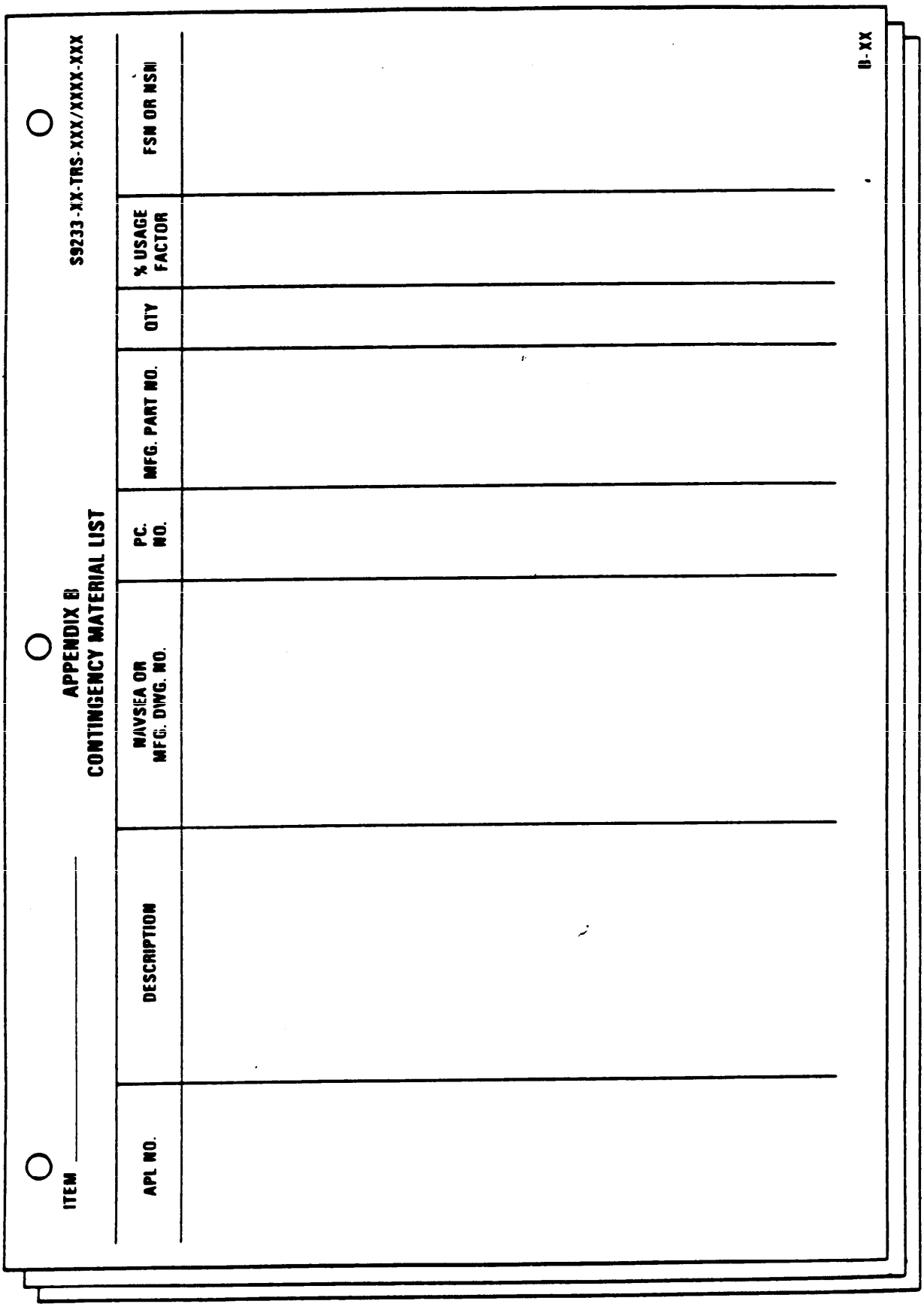


FIGURE 7-2. Contingency material list. (Example)

S9233-XX-TRS-XXX/XXXX-XXX-XXX

MATERIAL CONDITION OVERHAUL REPORT

SHIP:	HULL NO.

AS FOUND MATERIAL CONDITION (CHECK ONE)

☐ UNUSUALLY GOOD, MAINTENANCE COULD HAVE BEEN DELAYED	☐ AS EXPECTED, MAINTENANCE WAS NECESSARY	☐ UNUSUALLY BAD, MAINTENANCE SHOULD HAVE BEEN DONE SOONER

REMARKS:

AS OVERHAULED CONDITION RECORD

	YES	NO
WAS ALL MANDATORY OVERHAUL REPLACEMENT MATERIAL (APPENDIX A) REPLACED?		
COULD MANDATORY OVERHAUL REPLACEMENT MATERIAL HAVE BEEN REUSED? (EXPLAIN IF YES)		
DID OVERHAULED ITEM MEET ALL THE ACCEPTANCE CRITERIA IN THE TRS?		

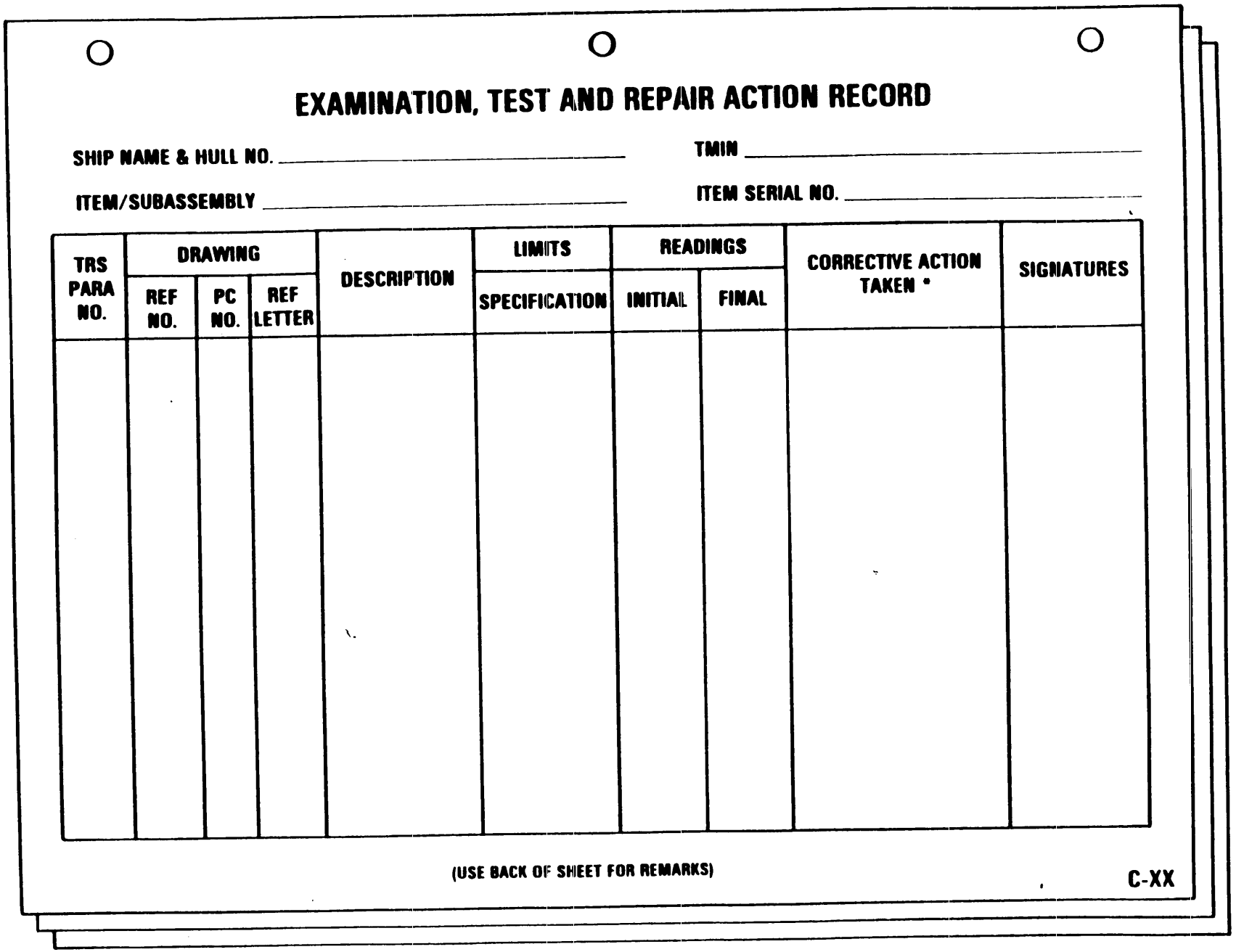
REMARKS: (EXPLAIN BLOCKS MARKED NO)

	YES	NO
WAS THERE AN APPROVED DEVIATION OF MATERIAL OR SPECIFIED PROCEDURES? (EXPLAIN IF YES)		

I CERTIFY, BASED ON PERSONAL KNOWLEDGE, THAT THE ABOVE ITEM HAS BEEN SATISFACTORILY OVERHAULED AND MEETS THE ACCEPTANCE CRITERIA PROVIDED HEREIN.

ORIGINAL 3-XX

FIGURE 7-3. Material condition/overhaul report.



EXAMINATION, TEST AND REPAIR ACTION RECORD

SHIP NAME & HULL NO. _____ TMIN _____

ITEM/SUBASSEMBLY _____ ITEM SERIAL NO. _____

TRS PARA NO.	DRAWING			DESCRIPTION	LIMITS	READINGS		CORRECTIVE ACTION TAKEN *	SIGNATURES
	REF NO.	PC NO.	REF LETTER		SPECIFICATION	INITIAL	FINAL		

(USE BACK OF SHEET FOR REMARKS)

C-XX

FIGURE 7-4. Examination test and repair action record.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-24784/7(SH)	2. DOCUMENT DATE (YYMMDD) 940906

3. DOCUMENT TITLE
MANUAL, TECHNICAL REPAIR STANDARDS (TRS) REQUIREMENTS FOR HM&E EQUIPMENT

4. NATURE OF CHANGE (identity paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

A. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

A. NAME Technical Point of Contact (TPOC) MR. MICKEY ANDER, SEA 04TD3 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN: TPOC: 703-602-8701 8-332-8701
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03R42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340